3.8 Networks using blocks (VGG)

- while AlexNet offered empirical evidence that deep CNNs can achieve good results, it did not provide a general template to guide subsequent researchers in designing new networks
- in the following sections, we will introduce several heuristic concepts commonly used to design deep networks
- progress in this field mirrors that in chip design, where engineers went from placing transistors to logical elements to logic blocks
- similarly, the design of neural network architectures had grown progressively more abstract, with researchers moving from thinking in terms of individual neurons to whole layers, and now to blocks, repeating patterns of layers
- the idea of using blocks first emerged from the Visual Geometry Group (VGG) at Oxford University, which gave the name of their *VGG* network
- it is easy to implement these repeated structures in code with any modern deep learning framework, by using loops and subroutines

3.8 Networks using blocks (VGG)

- the basic building block of classic CNNs is a sequence of the following:
  - a convolutional layer with padding to maintain the resolution
  - a nonlinearity such as a ReLU
  - a pooling layer such as a maximum pooling layer
- one VGG block consists of a sequence of convolutional layers, followed by a maximum pooling layer for spatial downsampling
- in the original VGG paper, the authors employed convolutions with 3×3 kernels with padding of 1 (keeping height and width) and 2×2 maximum pooling with stride of 2 (halving the resolution after each block)
- like AlexNet and LeNet, the VGG Network can be partitioned into two parts: the first consisting mostly of convolutional and pooling layers, and the second consisting of fully-connected layers; this is depicted in Figure 13

3.8 Networks using blocks (VGG)

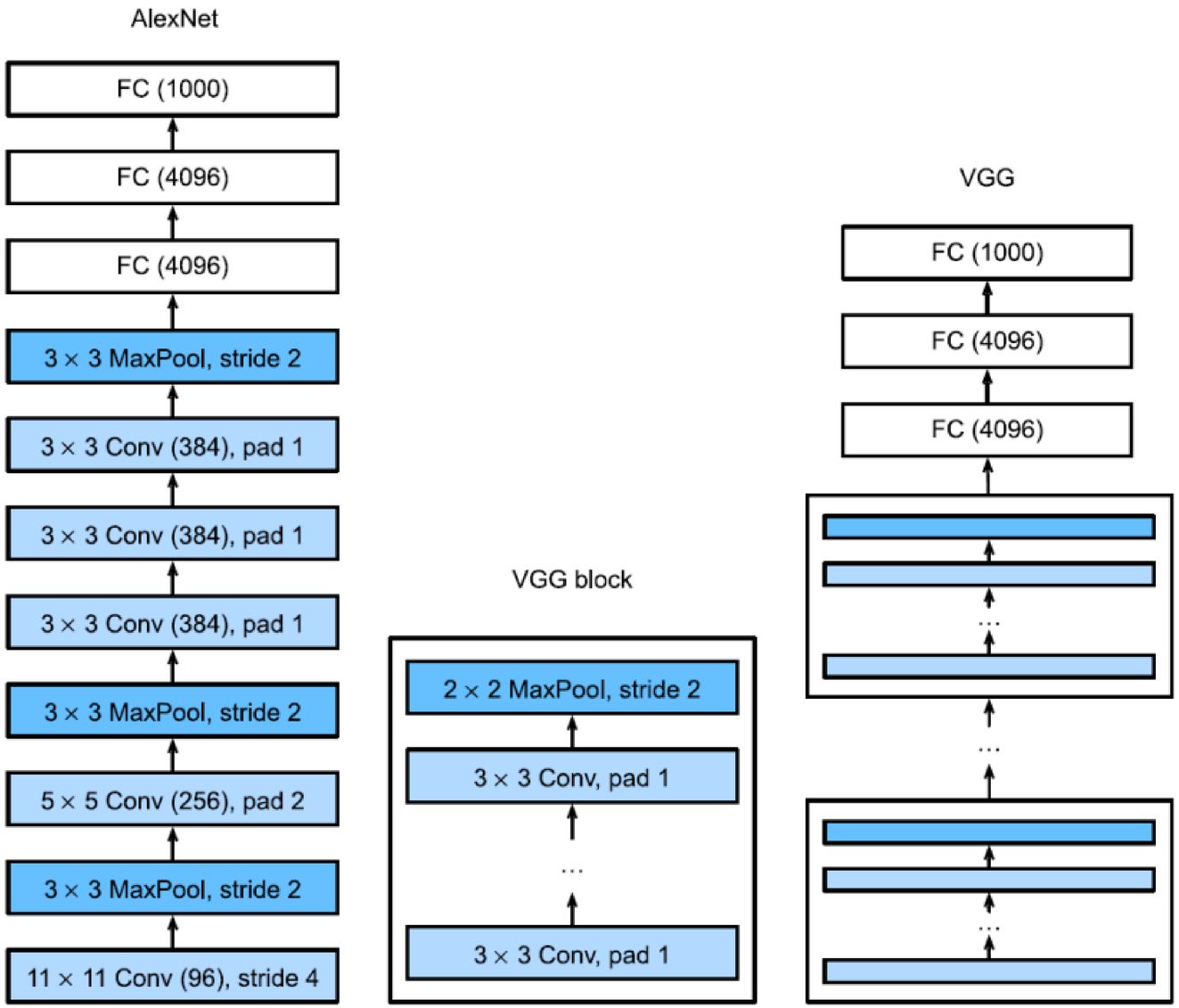

Figure 13: From AlexNet to VGG, which is designed from building blocks.

3.8 Networks using blocks (VGG)

- the convolutional part of the network connects several VGG blocks from Figure 13, in succession
- the fully-connected part of the VGG network is identical to that covered in AlexNet
- the original VGG network had 5 convolutional blocks, among which the first two have one convolutional layer each, and the latter three contain two convolutional layers each
- the first block has 64 output channels, and each subsequent block doubles the number of output channels, until that number reaches 512
- since this network uses 8 convolutional layers and 3 fully-connected layers, it is often called *VGG-11*